

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I D. Dhasane (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

D. Dhasane
Signature of the student:

Level-4
40

Dhareshana D1 192524243

1. A hospital needs to assign 3 nurses (N_1, N_2, N_3) wards (ICU, General, Emergency) such that.

Constraints:

- i N_1 cannot be assigned to emergency ward
- ii N_2 must be assigned before N_3
- iii Only one nurse per ward
- iv N_3 cannot be assigned to ICU
- v All nurses must be assigned exactly one ward

2. An autonomous rescue drone is deployed in a disaster-affected building to locate a survivor.
- constraints:

- i The drone can move Up, down, left, Right
- ii Some cells are blocked and cannot be traversed
- iii) The drone has limited sensing
- iv Each movement has cost
- v The drone must minimize total cost
- vi The drone must update its knowledge dynamically

Constraint
Search.

Constraint-based problem solving using backtracking Search.

Variables : N_1, N_2, N_3

Domains : $\{ICU, General, Emergency\}$

Constraints :

1. $N_1 \neq Emergency$
2. N_2 must be assigned to a ward that comes before N_3 in the order $ICU < General < Emergency$
3. Only one nurse per ward
4. $N_3 \neq ICU$
5. Each nurse gets exactly one ward

Step 1: Assign N_1

Possible wards : $ICU, General$

choose $N_1 = ICU$

Current assignment :

• $N_1 \rightarrow ICU$

Constraint check :

- $N_1 \neq Emergency$
- No conflicts

Step 2: Assign N_2

Remaining wards : $General, Emergency$

Try $N_2 = General$

Current assignment :

- $N_1 \rightarrow ICU$
- $N_2 \rightarrow General$

Constraint check:

- Unique wards

Step 3: Assign N3

Remaining ward: Emergency

Try N3 = Emergency

Current Assignment:

- N1 → ICU
- N2 → general
- N3 → Emergency

Constraint checks:

- N3 ≠ ICU
- N2 before N3
- One nurse per ward
- All nurses assigned

Search Tree

Start

1

N1 = ICU

1

N2 = General

1

N3 = Emergency

1

Solution Found

Final Valid Schedule

Nurse

Ward

N1

ICU

N2

General

N3

Emergency

2) Autonomous Rescue Drone Problem

a) Constraint Analysis.

1. Movement constraint

- Drone can move only Up, Down, Left, or Right
- Limits possible actions at each state

2. Obstacle constraint

- Blocked cells cannot be entered
- Reduces available paths.

3. Limited Sensing

- Drone knows only adjacent cells
- Environment is partially observable

4. Movement cost

- Normal move = 1
- Risky cell = $1 + 2 = 3$
- Encourages safer routes.

5. Cost Minimization

- Drone must choose the path with minimum total cost.

6. Dynamic knowledge Update

- As new cells are observed, the drone updates its map.
- Decisions are made online.

b) Solution using uniform cost search + Online

Example Grid

S = Start

G = Goal (Survivor)

X = Obstacle

R = Risky cell

S	.	R	.
.	X	.	.
.	.	X	.
R	.	.	G

Step 1: Start at S

known area contains only adjacent cells

Step 2: Expand lowest-cost Node.

Possible moves:

- Right (cost = 1)
- Down (cost = 1)

Choose any minimum-cost move.

Step 3: Update knowledge

Drone senses neighbouring cells and discovers:

- obstacles
- Risky zones
- Safe paths

Step 4: Continue UCS

Priority Queue stores nodes by cumulative cost.

Eg:

Path	Cost
S → Right	1
S → Down	1
S → Right → Right (Risky)	4
S → Down → Down	2

Lowest - cost node is always expanded first

Step 5: Reach Goal

The drone eventually finds a path such as
S → Down → Down → Right → Right → Down
G ←

avoiding risk cells where possible

c) AI concepts Applied

Intelligent Agent

The drone:

- Perceives environment through sensors
- Make decisions.
- Acts through movement.

Rationality :

A rational agent chooses actions that minimize total cost while maximizing safety

Environment type

Property	Type
Observability Agents changes during search Outcomes State Space	Partially observable Simple agent Dynamic Deterministic Discrete

- d) Why uniform cost search / online search is suitable
1. Handles different path costs effectively.
 2. Finds the minimum-cost path.
 3. Works well when risky zones have higher costs.
 4. Online search allows decision with incomplete information
 5. Supports dynamic map updates as new rooms are discovered.